

Plantlfi - Interactive Web Application Development
ICT 1209 – Web Technologies | Mini Project Final Report

Project Title: Plantlfi: An Interactive Botanical Database &
Safety Guide Theme: Botanical Directory (Scientific names, toxicity
profiling, edibility markers) Group Members:

I.A.V.K. Ilangakoon (Index No2732)

B.S.C. Hasarangi (Index No: 2729)

Date: September 7, 2026

2. Project Overview

Purpose: Plantifi solves the practical problem of making biological and botanical data accessible. It provides everyday users with a reliable platform to check plant toxicity, edibility, and economic value to prevent accidental poisonings and educate the public.

Target Audience: Casual gardeners, pet owners, agriculture students, landscape designers, and eco-tourists.

Key Features:

Dynamic Plant Directory: Instant client-side filtering by category (Edible, Toxic, Medicinal).

Interactive Carousel Slider: Highlighting featured plants and their safety warnings.

Secure User Authentication: PHP/MySQL powered registration and login system with password hashing.

Validated Inquiry Form: A contact form with dual validation (JavaScript frontend + PHP backend) to submit toxicity reports directly to the database

3. System Design

Table 1: users

Field Name	Data Type	Details
id	INT, AUTO INCREMENT	PRIMARY KEY
username	VARCHAR 50	NOT NULL UNIQU
email	VARCHAR 100	NOT NULL UNIQU
password	VARCHAR 255	NOT NULL
Created_at	TIMESTAMP	DEFAULT

Table 2: messages

Field Name	Data Type	Details
id	INT, AUTO INCREMENT	PRIMARY KEY
name	VARCHAR 100	NOT NULL
email	VARCHAR 100	NOT NULL
message	TEXT	NOT NULL
Created_at	TIMESTAMP	DEFAULT

Table 3: plants

Field Name	Data Type	Details
id	INT, AUTO INCREMENT	PRIMARY KEY
Common_name	VARCHAR 100	NOT NULL
Scientific_name	VARCHAR 100	NOT NULL
category	ENUM	NOT NULL
description	TEXT	
Image_url	VARCHAR 255	DEFAULT NULL
Toxicity_level	VARCHAR 50	DEFAULT

Folder Structure

-PlantIfi/

├── auth/

| ├── register.php

| ├── login.php

| └── logout.php

├── css/

| └── style.css

├── images/

├── includes/

| └── db.php

├── js/

| └── script.js

├── contact.php

├── dashboard.php

├── directory.php

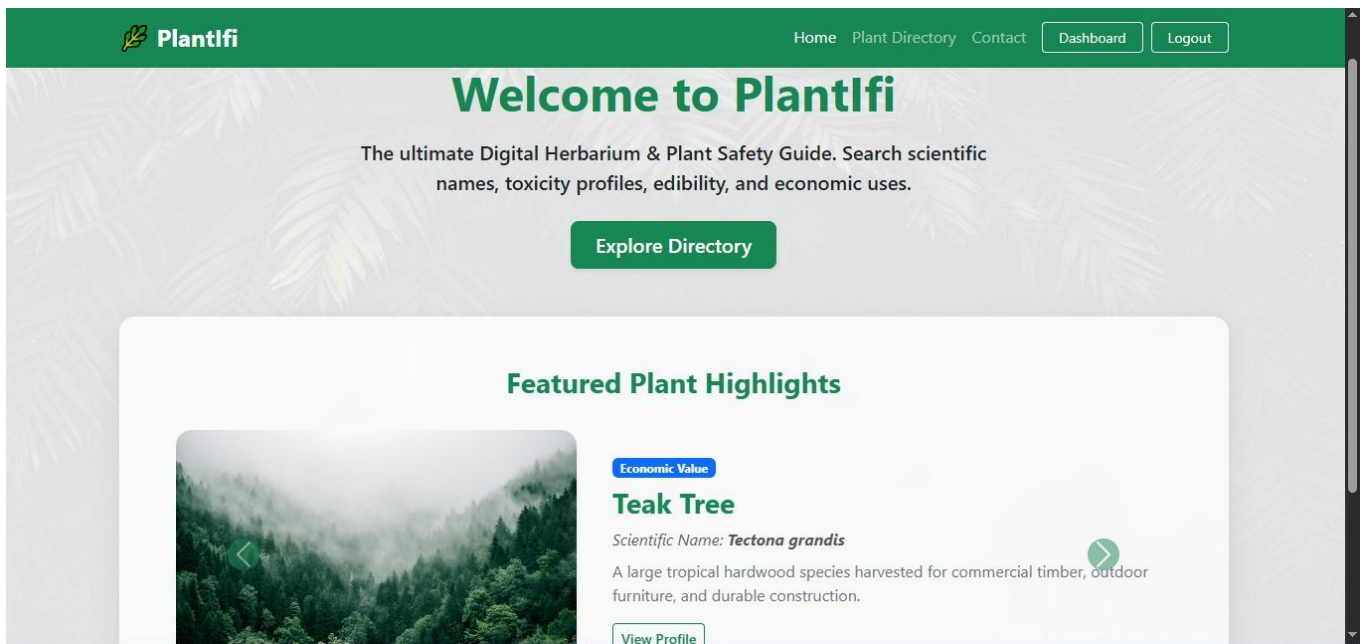
├── index.php

├── database.sql

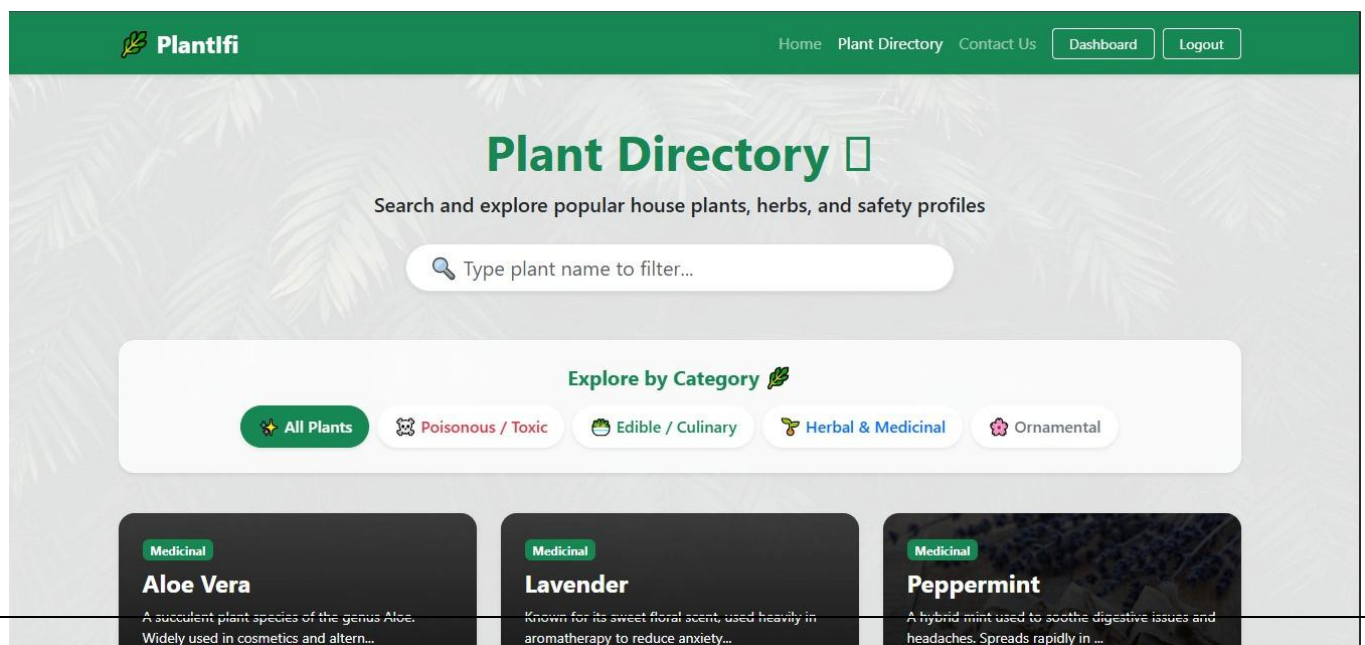
└── README.md

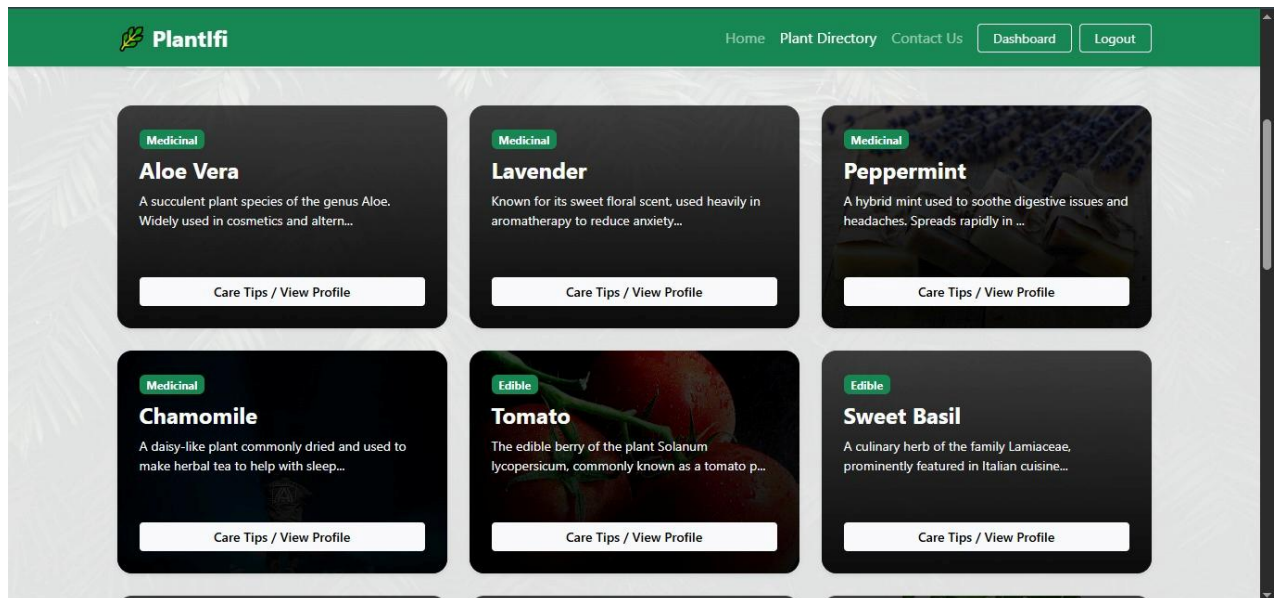
4. Implementation

Home Page (index.php) Responsive Bootstrap 5 layout with an interactive JavaScript carousel for featured plants.

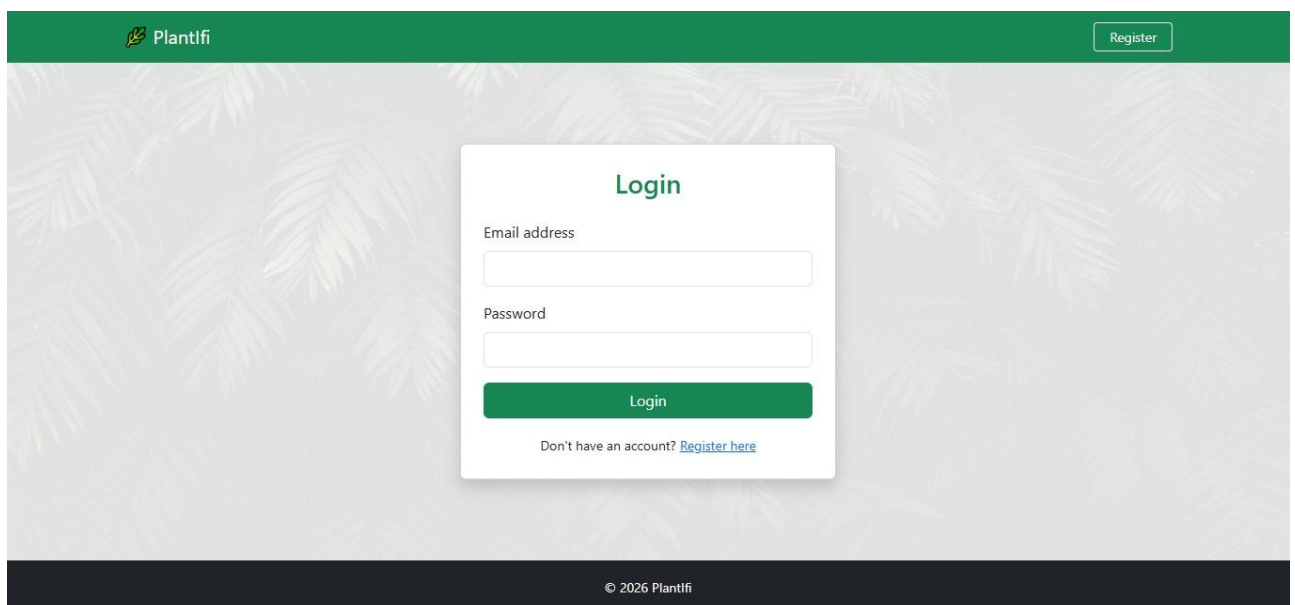


Plant Directory (directory.php) Fetches plants dynamically from the MySQL database and uses Vanilla JavaScript array filtering to instantly sort plants by category without reloading the page.





User Authentication (register.php & login.php) Implements PASSWORD_BCRYPT hashing for secure data storage. Prevents session fixation via session_regenerate_id().



Contact & Form Submission (contact.php) Captures user data, validates via regex in JS, and inserts it into the messages MySQL table using PHP PDO prepared statements to prevent SQL injection.

Home Plant Directory Contact Us Dashboard Logout

Support & Inquiries

Get in Touch with Plantifi

Have questions regarding botanical identification, plant toxicity, or safety guides?
Send our team a message below.

**Our Location**

Department of ICT, Rajarata
University of Sri Lanka, Mihintale

**Email Us**

support@plantifi.rus.ac.lk

Your plant toxicity report has been logged successfully!

Full Name

Email Address

Inquiry Type

phpMyAdmin

Server: 127.0.0.1 > Database: plantifi > Table: messages

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking Triggers

Showing rows 0 - 0 (1 total, Query took 0.0007 seconds.)

SELECT * FROM `messages`

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

	id	name	email	message	created_at
☐	1	Kamal Perera	kamal.p@gmail.com	Hi, I recently found a strange succulent in my gar...	2026-09-06 23:35:47

☐ Check all | With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

☐ Print ☐ Copy to clipboard ☐ Export ☐ Display chart ☐ Create view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Console

5. Challenges and Solutions

- Challenge: Preventing SQL Injection when users submit contact forms.

Solution: We implemented PDO Prepared Statements (`$pdo->prepare`) in PHP, which separates the SQL logic from the user input, making the database secure against injection attacks.

- Challenge: Maintaining front-end JS validation while submitting data to PHP.

Solution: We updated our JavaScript event listener to validate the form first. If the validation passes, the JS allows the native HTML form submission to trigger, passing the POST data perfectly to PHP.

6. Individual Contribution

Both team members contributed equally to the success of the project:

I.A.V.K. Ilangakoon (Index: 2732):

- Designed the frontend wireframes and UI layout (Bootstrap 5, CSS).
- Implemented the JavaScript filtering logic for the Plant Directory.
- Developed the interactive carousel on the Home Page.
- Created the database.sql schema and table structures.

B.S.C. Hasarangi (Index: 2729):

- Developed the PHP backend authentication (register.php, login.php, logout.php).
- Handled the database connection securely using PDO (includes/db.php).
- Wrote the PHP backend for the contact.php form to store messages in the database.
- Integrated the dynamic navigation bar across all PHP pages.

7. Setup Instructions

1. Install XAMPP and start the Apache and MySQL modules.
2. Open your browser and navigate to <http://localhost/phpmyadmin>.
3. Create a new database named plantifi.
4. Click the "Import" tab and upload the database.sql file provided in the source code folder.
5. Place the entire -PlantIfi project folder into your XAMPP htdocs directory (e.g., C:\xampp\htdocs\PlantIfi).
6. Access the application in your browser at <http://localhost/-PlantIfi/>.

8. Git repository link

<https://github.com/Basooriya/-PlantIfi>

9. References

- PHP Official Documentation: <https://www.php.net/docs.php>
- Bootstrap 5 Documentation: <https://getbootstrap.com/docs/5.3/>
- JavaScript MDN Web Docs: <https://developer.mozilla.org/en-US/docs/Web/JavaScript>